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Your grade: 100%

Your best: 100% · Your highest: 100%

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Module 3 Summative Assessment

Review Learning Objectives

Instructions

1. Which of the following is NOT a goal of early stopping in the context of deep learning?

☐

Preventing overfitting.

☐

Improving results on test data by fine-tuning.

☒

Increasing the number of epochs for training.

☐

Stabilizing the learning process.

Correct

Correct

Due

Dec 8, 12:59 PM IST

Attempts

1 left (2 attempts every 9 hours)

Submitted

Nov 22, 12:55:48 IST

Retry

1 / 1 point

2. What is the primary purpose of regularization in deep learning?

☐

To increase the size of the network.

☐

To minimize the original cost function only.

☒

To find a set of weights that not only minimizes the original cost function but also keeps weights close to zero.

☐

To maximize the performance on the training data only.

Correct

Correct

Like

Dislike

Report as issue

1 / 1 point

3. What is the primary reason for using Rectified Linear Unit (ReLU) in deep neural networks?

☐

It requires extensive computational resources.

☐

It helps in reducing the network's accuracy.

☒

It addresses the vanishing gradient problem.

☐

It increases the model's training time.

Correct

Correct

1 / 1 point

4. What is the third step in the recipe of deep learning?

☒

Pick the loss function.

☐

Define a set of function.

☐

Goodness of function.

☐

Modify the learning rate.

Correct

Correct

1 / 1 point

5. According to the slides, which statement best describes the role of a validation set in deep learning?

☐

It is used for final evaluation of the model's performance.

☒

It is used to adjust the model's hyperparameters based on its performance.

☐

It is part of the training data used to compute the gradient.

☐

It is used for training the model.

Correct

Correct

1 / 1 point

6. What analogy is used in the slides to explain the concept of weight decay in regularization?

☐

Trimming a tree to encourage healthy growth.

☒

Pruning out useless links between neurons, similar to forgetting unused information in the human brain.

☐

Adding more layers to a neural network to improve its capacity.

☐

Increasing the dataset size to improve model generalization.

Correct

Correct

1 / 1 point

7. What is one of the suggested solutions when achieving good results on training data but not on testing data?

☐

Increase the network size.

☐

Remove all regularization methods.

☒

Implement early stopping, regularization, and dropout.

☐

Always use a larger learning rate.

Correct

Correct

1 / 1 point

8. What problem does the Vanishing Gradient Problem describe?

☐

Gradients become too large, causing weights to update too rapidly.

☒

Gradients for early layers become very small, slowing down learning.

☐

The model immediately converges to a local minimum.

☐

Activation functions eliminate the gradient.

Correct

Correct

1 / 1 point

9. What is a sign of overfitting in the context of neural networks?

☒

Good results on training data but poor on testing data.

☐

Poor results on both training data and testing data.

☐

Good results on both training data and testing data.

☐

The model cannot converge during training.

Correct

Correct

1 / 1 point

10. What does the term "momentum" in gradient descent algorithms help to achieve?

☐

Decrease the learning rate.

☐

Increase the gradient.

☒

Help convergence in training.

☐

Guarantee reaching the global minimum.

Correct

Correct

1 / 1 point

11. Which regularization term provides a sparse solution?

☒

L-1 norm

☐

L-2 norm

☐

L-p,q norm

Correct

Correct

1 / 1 point

12. Which problem in deep learning trainings is described as being "very slow at the plateau"?

☒

Stuck at local minima

☐

Overfitting

☒

Stuck at saddle points

☐

Underfitting

Correct

Correct

1 / 1 point

13. What is the primary purpose of using dropout in deep learning?

☐

To increase the network's computational efficiency.

☒

To prevent overfitting by randomly dropping neurons during training.

☐

To enhance the network's ability to memorize training data.

☐

To increase the depth of the network without adding new neurons.

Correct

Correct

1 / 1 point

14. During the training phase, how is the dropout technique applied to a neural network?

☐

By randomly selecting a subset of neurons to be activated at each epoch.

☐

By permanently removing a percentage of neurons from the network.

☒

By randomly dropping a percentage of neurons in each mini-batch.

☐

By alternating the activation functions of neurons at each iteration.

Correct

Correct

1 / 1 point

15. How does the Rectified Linear Unit (ReLU) relate to Maxout in the context of activation functions in neural networks?

☐

ReLU and Maxout are entirely unrelated, serving different purposes in the network.

☐

ReLU is a precursor to Maxout, which has since become obsolete.

☒

ReLU is a special case of Maxout, illustrating a widely used approach to activation.

☐

Maxout is a simplified version of ReLU designed to reduce computational complexity.

Correct

Correct

1 / 1 point

https://www.coursera.org/learn/llnnois-1ech-deep-learning/assignment/submission/8f8eG7/module-3-summative-assessment/view-feedback

1/1